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Bereken $\int \cosh^3 x dx$

$$\begin{aligned}\int \cosh^3 x dx &= \int \cosh x \cdot \cosh^2 x dx \\ &= \int \cosh x (1 + \sinh^2 x) dx\end{aligned}$$

Substitueer $t = \sinh x$, dan is $dt = \cosh x dx$

$$\begin{aligned}&= \int (1 + t^2) dt \\ &= t + \frac{t^3}{3} + c \\ &= \sinh x + \frac{\sinh^3 x}{3} + c\end{aligned}$$

References